UPSTACK · PROJECT 01

UpStack

AI-Powered API Monitoring and Incident Management System

Developed using Zoho Creator



Student name: [Your Name] · College: [College Name] · Department: [Department]

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THE VISIBILITY GAP

API failures are hard to see early

API monitoring checks whether services are available and responding within an acceptable time.



Connected apps slow down

Failures can make connected applications slow or unavailable.

Manual checks miss moments

Manual checking takes time and may miss short incidents.



One place changes the response

UpStack brings health checks, failure detection, AI analysis, and incident tracking into one place.

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THE PROBLEM

The problem: failures become incidents before teams can act

- APIs may become unavailable without a clear warning.
- Slow response times affect the applications that depend on them.
- Manual monitoring is repetitive and difficult to scale.
- Failure detection and incident tracking are often handled separately.

Project focus: make the first response faster and easier to follow.

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THE CORE WORKFLOW

UpStack turns monitoring into a simple workflow

Each step moves from checking an API to making the result visible and actionable.

01

Monitor

Monitor API availability.

02

Measure

Measure API response time.

03

Detect

Detect API failures.

04

Analyse

Analyse errors using AI.

05

Manage

Automatically create and manage incidents.

06

Display

Display monitoring information through a dashboard.

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IMPLEMENTATION STACK

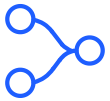
Built on Zoho Creator with focused integrations

Only the tools used by the implemented project are shown here.



Zoho Creator

Low-code application platform.



Deluge

Scripting for workflows and logic.



Zia AI

AI-powered error analysis.



OpenAI and Gemini

Additional AI integrations.



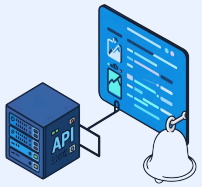
Email notifications

Failure alerts for users.

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IMPLEMENTED ARCHITECTURE

From API check to dashboard visibility



API Endpoints

API Health Monitoring

API Check History

Failure Detection

Zia AI Analysis

Incident Creation

Email Notification

Dashboard

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SYSTEM STRUCTURE

Four modules keep the system understandable



API Endpoints

Stores API details.



API Check History

Stores status, response time, and errors.



Incident Management

Tracks and manages failures.



Dashboard

Displays monitoring statistics.

The modules separate configuration, evidence, response, and visibility.

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Workflow and implementation

The workflow connects an external API call to an incident record and an alert.

01

Register

Register an API endpoint.

02

Check

Perform an API health check.

03

Store

Store the response status and time.

04

Detect

Detect failures.

05

Analyse

Analyse errors using Zia AI.

06

Respond

Create an incident and send an email alert.

Implementation notes: Deluge workflows coordinate the steps, and invokeurl supports external API calls.

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VISIBILITY AT A GLANCE

The dashboard makes API health visible

12 · Total API Endpoints

08 · Total Incidents

420 ms · Average Response Time

05 · Failed Health Checks

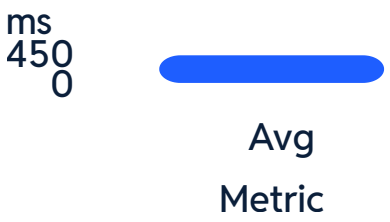
03 · Open Incidents

94% · Healthy API Checks

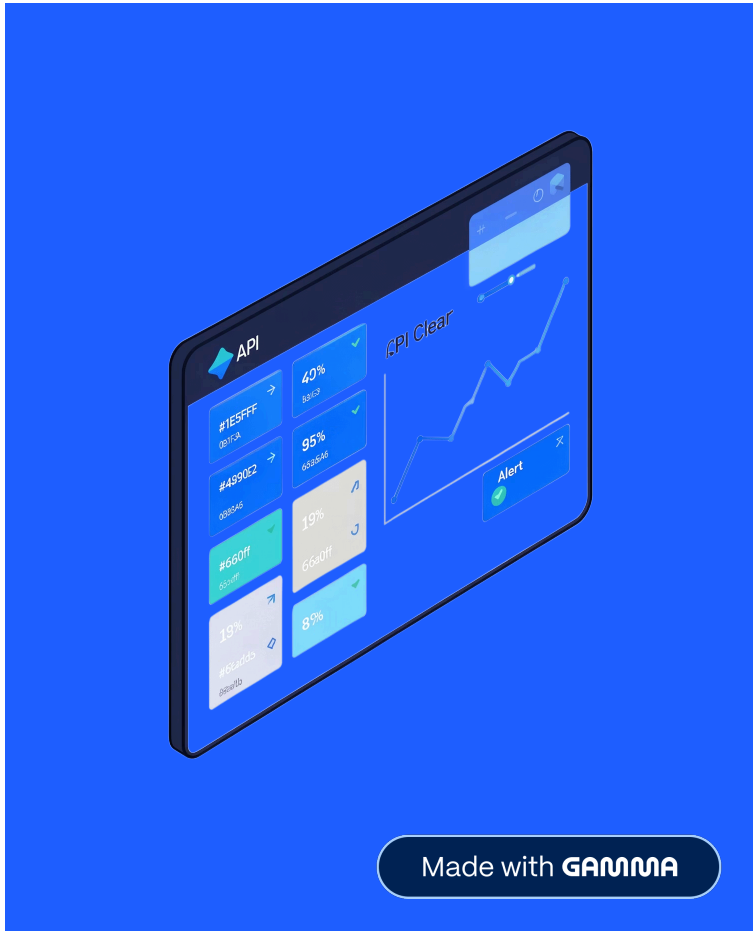
Incident status · illustrative



Average response time · illustrative



All figures on this slide are illustrative values for the presentation.



CLOSING

Conclusion and future enhancements

IMPLEMENTED

UpStack simplifies API monitoring

It detects failures, uses AI for error analysis, and improves incident management.

FUTURE

Extend the monitoring experience

Future enhancements include more API providers, custom monitoring intervals, advanced analytics, and an improved alerting system.

Thank You – Questions?

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